CURRICULUM VITAE



Dr. Danny Fainozzi

Physicist · ultrafast EUV and X-ray science

Advanced Postdoctoral Researcher, Institute of Applied Physics, University of Bern

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I develop nonlinear X-ray spectroscopies that watch heat, sound and spin move through matter on the nanometre and femtosecond scale.

25+ journal articles

5 X-ray FEL facilities

6 nm smallest transient grating pitch

10+ yrs at large-scale facilities

01 Employment

03.2024 – present — **Advanced Postdoctoral Researcher**
University of Bern, CH · Institute of Applied Physics
Prof. Andrea Cannizzo

04.2021 – 03.2024 (3 years) — **Junior Postdoctoral Researcher**
Elettra Sincrotrone Trieste, IT · FERMI Free-Electron Laser, TIMER endstation
Dr. Filippo Bencivenga

11.2017 – 10.2020 (3 years) — **Doctoral Researcher**
Elettra Sincrotrone Trieste, IT · FERMI Free-Electron Laser (PhD, University of Trieste)
Dr. Claudio Masciovecchio

02 Education

11.2017 – 10.2020 (3 years) — **PhD / Dr.: Physics**
University of Trieste, IT · Elettra Sincrotrone Trieste; thesis in collaboration with the Paul Scherrer Institute, CH
Advisor: Dr. Claudio Masciovecchio

10.2015 – 07.2017 (1 year 10 months) — **Master: Condensed Matter Physics**
University of Trieste, IT · Thesis in collaboration with the Kyoto Institute of Technology, JP
Advisor: Prof. Fulvio Parmigiani

10.2012 – 07.2015 (2 years 10 months) — **Bachelor: Physics**
University of Trieste, IT · Thesis in collaboration with Elettra Sincrotrone Trieste
Advisor: Prof. Fulvio Parmigiani

03 Research stays abroad

2 months — **Visiting researcher**
Massachusetts Institute of Technology, US · Optical Faraday-rotation measurements of nanoscale spin waves
Host: Prof. Keith A. Nelson

2017 — **Master thesis research**
Kyoto Institute of Technology, JP · Master thesis project
Host: Prof. Giuseppe Pezzotti

04 Large-scale facilities

More than ten years of experiments with femtosecond EUV and X-ray free-electron lasers, plus ultrafast optical tabletop setups.

FERMI FEL (Elettra, IT) — TIMER endstation
EUV transient gratings and EUV imaging: proposer, setup preparation, experiments, data analysis

SwissFEL (PSI, CH)
First hard X-ray transient grating (X-ray pump / optical probe): preparation, experiment, data analysis

European XFEL (DE) — FXE, MID
Long-term proposal for X-ray transient gratings: shift leader and lead data analyst

SACLA (JP)
All hard X-ray transient gratings: experiment and data analysis; campaign coordination

LCLS (SLAC, US)
All-X-ray transient grating campaign: coordination

05 Funding

CHF 400,000 — **SNSF COST project cooperation, linked to COST Action CA22148 (NEXT)**
Swiss National Science Foundation · University of Bern, CH
Awarded together with Prof. Andrea Cannizzo

06 Major achievements

01 Nanoscale acoustics and heat with EUV transient gratings

In a transient grating (TG) experiment two time-coincident pump pulses of wavelength λ cross on the sample at an angle 2θ and write a spatial modulation of excitation with period Λ = λ/(2 sin θ); a delayed probe pulse diffracts from it and reads out how it evolves. Free-electron lasers extend this approach into the extreme ultraviolet [17], giving contactless access to thermoelastic dynamics at length scales no optical laser can reach.

At the TIMER endstation of the FERMI FEL I acted as proposer, setup preparator, experimentalist and data analyst, and wrote the analysis codes shared with collaborators and students. I demonstrated stimulated Brillouin scattering in the time domain at a 1 nm^{-1} wave vector, an equivalent TG period of 6 nm, the smallest pitch recorded to date [14], and tracked nanoscale heat propagation through the thermoelastic response of a SiO_2 membrane [12]. The same platform gave nanoscale polarization gratings [6], time-domain EUV diffuse-scattering spectroscopy of surface phonons [4], and the electron-lattice coupling and bilayer thermoelasticity of metal films [1, 2].

02 Pioneering hard X-ray transient gratings

I contributed to the first hard X-ray transient grating, an X-ray pump / optical probe proof of principle on bismuth germanate at SwissFEL [19], supported by heterodyne-detected optical TG data [15], with responsibility for preparation, execution and data analysis. I then wrote the experimental protocols for X-ray TG at FELs[8], measured X-ray circular dichroism by cross-polarized X-ray TG [7], and contributed to the theory of soft X-ray gratings [11].

The all hard X-ray transient grating (X-ray pump / X-ray probe) was demonstrated at SACLA [5], where I prepared and ran the experiment and analysed the data; this milestone was a pivot for the COST Action CA22148. Hard X-ray gratings have since launched coherent spin waves [3]. The technique is developed under a long-term proposal at the European XFEL, where I am shift leader and lead data analyst at FXE and MID, and I coordinated campaigns at SACLA and LCLS.

03 Single-shot three-dimensional EUV imaging

I demonstrated stereoscopic three-dimensional coherent diffraction imaging with a single EUV FEL pulse, reaching sub-micrometre resolution [16], as experimentalist and data analyst. Single-shot 3D imaging opens a route to nanostructures in radiation-sensitive samples such as molecules, bacteria and viruses, without averaging over many identical copies.

04 Spin waves, phonons and EUV optics

In collaborative EUV TG campaigns I contributed to the excitation and detection of coherent nanoscale spin waves, supported by optical Faraday-rotation data I took during a research stay in Keith Nelson's group at MIT [9]; to understanding how echo interference controls phonon attenuation in nanophononic membranes [13]; to a high-efficiency programmable modulator for EUV light [10]; and to element- and enantiomer-selective real-time visualization of molecular motion [18].

07 Collaborations, leadership and service

- Member of COST Action CA22148 (NEXT, nonlinear EUV to hard X-ray techniques) since its start; the all-X-ray transient-grating demonstration was a pivot for its creation.

- Shift leader and lead data analyst in the European XFEL long-term proposal developing X-ray transient gratings at FXE and MID.
- Coordinator of experimental campaigns at SACLA and LCLS to advance all-X-ray transient gratings.
- Author of custom analysis codes for FEL four-wave-mixing data, shared with collaborators and students and open-sourced whenever possible.

08 Publications

19 selected peer-reviewed articles and preprints. FIRST AUTHOR marks first-author works; numbers match the references above. Full list on ORCID.

2026

1. **Ultrafast electron-lattice coupling in cobalt films probed by nanoscale extreme ultraviolet transient gratings**
   Kurdi, G., **Fainozzi, D.**, Foglia, L., Khatu, N., Laterza, S., Mincigrucci, R., … Masciovecchio, C., & Bencivenga, F.
   Physical Review B 114 · doi:10.1103/j1rk-j6rt

2. **Thermoelastic dynamics of a nanoscale metallic film: role of bilayer coupling revealed by extreme ultraviolet transient gratings**
   Krstulović, M., Pelli-Cresi, J. S., **Fainozzi, D.**, Foglia, L., Griesi, A., … Wehinger, B., & Bencivenga, F.
   Journal of Applied Physics 140, 065102 · doi:10.1063/5.0340562

3. **Spin waves excited by hard X-ray transient gratings**
   Miedaner, P. R., Maznev, A. A., Biednov, M., … **Fainozzi, D.**, … Cannizzo, A., Milne, C., Nelson, K. A., & Svetina, C.
   Physical Review Letters · doi:10.1103/zybx-dqfc

2025

4. **Time-domain extreme-ultraviolet diffuse scattering spectroscopy of nanoscale surface phonons**
   Capotondi, F., Maznev, A. A., Bencivenga, F., Bonetti, S., Engel, D., **Fainozzi, D.**, … & Trigo, M.
   Physical Review Letters 135, 266101 · doi:10.1103/dq8w-62bm

5. **All hard X-ray transient grating spectroscopy**
   Ferrari, E., Ueda, H., **Fainozzi, D.**, Osaka, T., Bencivenga, F., … & Svetina, C.
   Communications Physics 8 · doi:10.1038/s42005-025-02178-y

2024

6. **Nanoscale polarization transient gratings**
   Foglia, L., Wehinger, B., Perosa, G., Mincigrucci, R., … **Fainozzi, D.**, … Masciovecchio, C., Bonetti, S., & Bencivenga, F.
   Nature Communications 15 · doi:10.1038/s41467-024-54799-6

7. **X-ray circular dichroism measured by cross-polarization X-ray transient grating**
   Rouxel, J. R., Mincigrucci, R., **Fainozzi, D.**, & Masciovecchio, C.
   Journal of Physics B: Atomic, Molecular and Optical Physics · doi:10.1088/1361-6455/ad6b62

8. **Protocols for X-ray transient grating pump/optical probe experiments at X-ray free electron lasers** FIRST AUTHOR
   Fainozzi, D., Cucini, R., Vila-Comamala, J., Lima, F., … & Svetina, C.
   Journal of Physics B: Atomic, Molecular and Optical Physics · doi:10.1088/1361-6455/ad717f

9. **Excitation and detection of coherent nanoscale spin waves via extreme ultraviolet transient gratings**
   Miedaner, P. R., Berndt, N., Deschamps, J., **Fainozzi, D.**, … & Maznev, A. A.
   Science Advances 10(36), eadp6015 · doi:10.1126/sciadv.adp6015

10. **A high-efficiency programmable modulator for extreme ultraviolet light with nanometre feature size based on an electronic phase transition**
    Vaskivskyi, I., Mraz, A., Venturini, R., … **Fainozzi, D.**, Bencivenga, F., Masciovecchio, C., & Mihailovic, D.
    Nature Photonics · doi:10.1038/s41566-024-01389-z

11. **A new framework for soft X-ray transient gratings**
Foglia, L., Mincigrucci, R., Döring, F., **Fainozzi, D.**, Mamyrbayev, T., Masciovecchio, C., & Bencivenga, F.
Journal of Physics B: Atomic, Molecular and Optical Physics · doi:10.1088/1361-6455/ad4fd0

12. **Temperature effects on the nanoscale thermoelastic response of a SiO_2 membrane** FIRST AUTHOR
Fainozzi, D., Caporaletti, F., Capotondi, F., De Angelis, D., … Bencivenga, F., & Baldi, G.
APL Materials 12(5) · doi:10.1063/5.0201769

13. **The effect of echoes interference on phonon attenuation in a nanophononic membrane**
Hadi, M., Luo, H., Pailhès, S., Tanguy, A., … **Fainozzi, D.**, … & Giordano, V. M.
Nature Communications 15, 1317 · doi:10.1038/s41467-024-45571-x

14. **Stimulated Brillouin scattering in the time domain at 1 nm^{-1} wave vector** FIRST AUTHOR
Fainozzi, D., Foglia, L., Khatu, N. N., Masciovecchio, C., Mincigrucci, R., Paltanin, E., & Bencivenga, F.
Physical Review Letters 132, 033802 · doi:10.1103/PhysRevLett.132.033802

2023

15. **BGO relaxation dynamics probed with heterodyne detected optical transient gratings** FIRST AUTHOR
Fainozzi, D., Catalini, S., Torre, R., Masciovecchio, C., & Svetina, C.
arXiv:2309.02308 · preprint · doi:10.48550/arXiv.2309.02308

16. **Three-dimensional coherent diffraction snapshot imaging using extreme-ultraviolet radiation from a free electron laser** FIRST AUTHOR
Fainozzi, D., Ippoliti, M., Bille, F., De Angelis, D., … Bencivenga, F., & Capotondi, F.
Optica 10(8), 1053–1058 · doi:10.1364/OPTICA.492730

17. **Extreme ultraviolet transient gratings: a tool for nanoscale photoacoustics**
Foglia, L., Mincigrucci, R., Maznev, A. A., Baldi, G., … **Fainozzi, D.**, … & Bencivenga, F.
Photoacoustics 29, 100453 · doi:10.1016/j.pacs.2023.100453

18. **Element- and enantiomer-selective visualization of molecular motion in real-time**
Mincigrucci, R., Rouxel, J. R., Rossi, B., … **Fainozzi, D.**, … Chergui, M., & Masciovecchio, C.
Nature Communications 14, 386 · doi:10.1038/s41467-023-36047-5

2021

19. **Hard X-ray transient grating spectroscopy on bismuth germanate**
Rouxel, J. R., **Fainozzi, D.**, Mankowsky, R., Rösner, B., … & Svetina, C.
Nature Photonics 15, 499–503 · doi:10.1038/s41566-021-00797-9